PROBABILITY AND STASTICTICS – FOUNDATIONAL

Directions for questions 1 to 3: Select the correct alternative from the given choices.

A coin is tossed four times. Find the probability of getting exactly two heads.

- a) $\frac{1}{4}$
- b) $\frac{3}{8}$ (Your answer - ✓)
- c) $\frac{7}{16}$
- d) $\frac{5}{16}$

Solution:

The two heads can occur in the first and the second tosses, the first and the third tosses, the first and the fourth tosses, the second and the third tosses, the second and the fourth tosses or the third and the fourth tosses. There are 6 favourable ways.

Alternately, number of favourable ways = Number of ways of choosing the two tosses which should show heads = ${}^4C_2 = 6$.

Required probability = $\frac{6}{2^4} = \frac{3}{8}$. Choice (B)

Directions for questions 1 to 3: Select the correct alternative from the given choices.

A dice is rolled five times. Find the probability of getting a prime number exactly four times.

- a) $\frac{5}{16}$
- b) $\frac{5}{64}$
- c) $\frac{5}{32}$ (Your answer - ✓)
- d) $\frac{5}{128}$

Solution:

The prime numbers are 2, 3 and 5. Probability (Getting a prime number in a roll) $= \frac{3}{6} = \frac{1}{2}$.

Probability (Getting a non-prime number in a roll) = 1 – Probability (Getting a prime number in a roll) $= 1 - \frac{1}{2} = \frac{1}{2}$.

A non-prime number can occur in any of the five rolls.

Required probability $= 5\left(\frac{1}{2}\right)^4\left(\frac{1}{2}\right) = \frac{5}{32}$. Choice (C)

Directions for questions 1 to 3: Select the correct alternative from the given choices.

A card is picked from a well-shuffled pack of cards. Find the probability that the card is not a honour card (A, J, Q or K).

- a) $\frac{7}{13}$
- b) $\frac{5}{13}$
- c) $\frac{11}{13}$
- d) $\frac{9}{13}$

Solution:

There are 16 honour cards (A, J, Q and K) in any pack of cards.

Probability of the card picked being a non-honour card = $1 - \frac{16}{52} = \frac{9}{13}$. Choice (D)

Directions for questions 4 and 5: Type in your answer in the input box provided below the question.

There are three red balls and four green balls in a bag and five red balls and six green balls in another bag. A bag was chosen at random and a ball was randomly drawn from it. Find the probability that the ball is red.

Solution:

Probability (Choosing any of the two bags) = $\frac{1}{2}$.

Required probability = $\frac{1}{2}\left(\frac{3}{7}\right) + \frac{1}{2}\left(\frac{5}{11}\right) = \frac{1}{2}\left(\frac{68}{77}\right)$

$= \frac{34}{77}$.

Ans : (34)

Directions for questions 4 and 5: Type in your answer in the input box provided below the question.

A number is chosen randomly from the first 100 natural numbers. Find the probability that the number is a multiple of either 11 or 13.

Solution:

In the first 100 natural numbers, there are 9 multiples of 11 and 7 multiples of 13. Any multiple of both 11 and 13 must be a multiple of the LCM of 11 and 13, i.e. of 143. There is no multiple of 143 in the first 100 natural numbers. So, there is no multiple of both 11 and 13 in the first 100 natural numbers.

Number of multiples of either 11 or 13 = Number of multiples of 11 + Number of multiples of 13 – Number of multiples of both 11 and 13
= 9 + 7 – 0 = 16.

Required probability = $\frac{16}{100}$. Ans : (0.16)

Directions for questions 6 to 9: Select the correct alternative from the given choices.

Two dice are rolled together. Find the probability that the sum of the numbers appearing on them is more than 10.

- a) $\frac{1}{18}$
- b) $\frac{1}{12}$
- c) $\frac{1}{9}$
- d) $\frac{5}{36}$

Solution:

The maximum sum of the numbers appearing on the two dice is 12. (This occurs when each die has the maximum number appearing on it, i.e., when each die has 6 appearing on it)
Possible sums are 11 and 12.
If the sum is 11, the numbers on the dice are 5 and 6 or 6 and 5.
If the sum is 12, the numbers on the dice are 6 and 6.

Required probability = $3\left(\frac{1}{6}\right)\left(\frac{1}{6}\right) = \frac{1}{12}$. Choice (B)

Directions for questions 6 to 9: Select the correct alternative from the given choices.

Three coins are tossed together. Find the probability that exactly two coins show tails.

- a) $\frac{3}{8}$
- b) $\frac{1}{8}$
- c) $\frac{1}{4}$
- d) $\frac{1}{2}$

Solution:

The favourable outcomes are TTH, HTT, THT.
Total number of outcomes is 2^3, i.e. 8.
Required probability $= \frac{3}{8}$. Choice (A)

Directions for questions 6 to 9: Select the correct alternative from the given choices.

Two cards are drawn from a well-shuffled pack of cards. Find the probability that both are numbered cards of the same colour.

- a) $\frac{342}{52_{C_2}}$
- b) $\frac{324}{52_{C_2}}$
- c) $\frac{306}{52_{C_2}}$
- d) None of these

Solution:

Let us assume that both the cards are black. Number of ways of them being numbered cards = ${}^{18}C_2$.
Let us now assume that both the cards are red. Number of ways of them being numbered cards = ${}^{18}C_2$.
Required probability $= \frac{18_{C_2} + 18_{C_2}}{52_{C_2}}$

$= \frac{2 \times 18_{C_2}}{52_{C_2}} = \frac{306}{52_{C_2}}$. Choice (C)

Directions for questions 6 to 9: Select the correct alternative from the given choices.

Two cards are drawn from a well-shuffled pack of cards. Find the probability that either both are kings, or both are black.

- a) $\frac{330}{52_{C_2}}$
- b) $\frac{335}{52_{C_2}}$
- c) $\frac{340}{52_{C_2}}$
- d) $\frac{331}{52_{C_2}}$

Solution:

Required probability = Probability (Both cards are kings) + Probability (Both cards are black) – Probability (Both cards are black kings) $= \frac{{}^{4}C_2 + {}^{26}C_2 - {}^{2}C_2}{{}^{52}C_2} = \frac{330}{{}^{52}C_2}$

Choice (A)

Directions for question 10: Type in your answer in the input box provided in the question.

In the month of February of a non-leap year, the probability that there are five Mondays is

Your Answer: 0 ✅

Solution:

In the month of February of a non-leap year, each day of the week occurs exactly four times.

∴ Required probability = 0. Ans : (0)

Directions for questions 11 and 12: Select the correct alternative from the given choices.

If the arithmetic mean of 29, 32, 37, 44, 31 and x is 36, find the value of x.

- a) 47
- b) 41
- c) 49
- d) 43 (Your answer - ✅)

Solution:

Arithmetic mean $= \frac{29 + 32 + 37 + 44 + 31 + x}{6} = 36.$

$173 + x = 216.$

$x = 43.$ Choice (D)

Directions for questions 11 and 12: Select the correct alternative from the given choices.

If the median and the mean of a set of observations of a moderately symmetric distribution are 14 and 15, find their mode.

- a) 12 (Your answer - ✅)
- b) 11
- c) 8
- d) 13

Solution:

For a moderately symmetric distribution, mode = 3median – 2mean.
Mode = 3(14) – 2(15) = 12. Choice (A)

Directions for questions 13 and 14: Type in your answer in the input box provided in the question.

The harmonic mean of 7 and 9 is

Solution:

The harmonic mean of two numbers a and b is $\frac{2(a)(b)}{a+b}$.

The harmonic mean of 7 and 9 is $\frac{2(7)(9)}{7+9} = \frac{63}{8} = 7.875$. Ans : (7.875)

Directions for questions 13 and 14: Type in your answer in the input box provided in the question.

The geometric mean of 4, 8 and 16 is

Your Answer: 8 ✅

Solution:

The geometric mean of $a_1, a_2, a_3 \ldots\ldots\ldots a_n$

$= \sqrt[n]{a_1 \cdot a_2 \cdot a_3 \ldots \ldots \ldots . a_n}$

The geometric mean of 4, 8 and 12 = $\sqrt[3]{4 \cdot 8 \cdot 16} = \sqrt[3]{512} = 8$. Ans : (8)

Directions for question 15: Type in your answer in the input box provided below the question.

If the arithmetic mean and the geometric mean of two numbers are 24 and 12 respectively, find their harmonic mean.

Your Answer: 6 ✅

Solution:

If a and b are two positive numbers, $[\mathrm{GM}(a, b)]^2 = \mathrm{AM}(a, b) \times \mathrm{HM}(a, b)$.
Let the harmonic mean of the two numbers be x.
$12^2 = 24 \cdot x \Rightarrow x = 6$. Ans : (6)